VANDEWIELE

ROMANIA

NPI Module
User Manual

New Product Introduction — Project Management System

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Author:	GTMC — Gianluca Testa
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1. Introduction to NPI

1.1 What is NPI?

NPI (New Product Introduction) is a structured process for managing the introduction of new products into production. The NPI module provides a comprehensive project management system that tracks every phase of the process — from initial design through to final production approval.

Each NPI project is associated with a product and consists of multiple waves, categories, and tasks. Tasks are assigned to responsible persons (owners) and tracked with start dates, due dates, and completion status.

1.2 Module Overview

Feature	Description
Dashboard	Central overview of all NPI projects with KPI statistics
Setup	Configuration of subjects, products, categories, and task templates
Project Detail	Full task management with dependencies, documents, and cost tracking
Gantt Chart	Interactive timeline visualization with Plotly (standard + consolidated)
Analysis	Overdue task analysis with manual email reminder capability
Hierarchy	Parent-child project relationships for complex product variants
Auto Notifications	Scheduled email alerts for overdue and expiring tasks

2. Initial Setup

Before creating NPI projects, you must configure the foundational data. Access the setup via the main menu: **NPI** → **Configuration**.

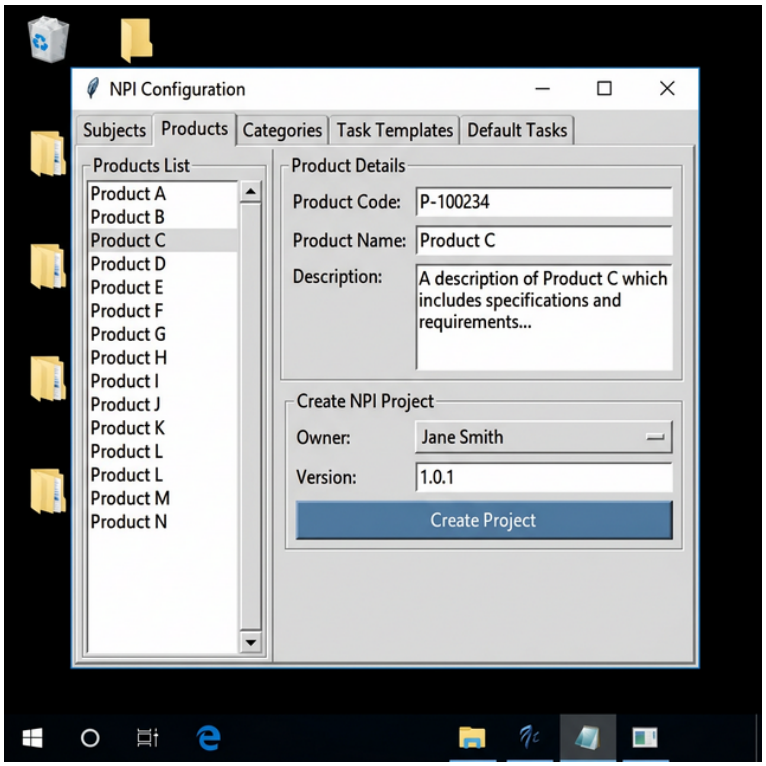

Figure 1 — NPI Configuration Window (Products tab)

2.1 Subjects (People & Contacts)

Subjects represent the people involved in NPI projects. Each subject has a Name, Email, Role, and Type (Internal, Customer, or Supplier). Only **Internal** subjects can be assigned as task owners and receive email notifications.

Steps:

1. Select the **Subjects** tab
2. Click **New** to create a new subject or select an existing one from the list
3. Fill in Name, Email, Role, and Type fields
4. Click **Save** to confirm
5. Use **Delete** to remove unused subjects

> **Tip:** The system can automatically link subjects to the HR database by matching names. External subjects (customers and suppliers) are excluded from automated notifications.

2.2 Products

Products are the items being introduced. Each product has a Code, Name, and Description. An NPI Project is always created starting from a product.

Creating a Product:

1. Select the **Products** tab

2. Enter Product Code and Product Name
3. Click **Save**

Creating an NPI Project from a Product:
1. Select the product in the list
2. In the **Create NPI Project** section, select the Owner and Version
3. Optionally attach initial documents
4. Click **Create Project**

Warning: Deleting a product performs a soft-delete (DateOut is set). Products with active projects cannot be deleted. Use the **Restore** button to recover deleted products.

2.3 Categories & Task Templates

Categories group tasks logically (e.g., Design, Validation, Production Setup). Each category has a sort order (NrOrdin) that determines display sequence.

Task Templates are pre-defined tasks within each category. When a new NPI project is created, tasks are automatically generated from the templates marked as **Default**.

2.4 Default Tasks Configuration

The **Default Tasks** tab allows you to define which task templates are automatically included when creating new NPI projects. Tasks marked as default will be pre-populated with their category and wave assignments.

3. NPI Dashboard

The Dashboard is the main entry point for the NPI module. It provides a real-time overview of all active projects with filtering, statistics, and quick actions.

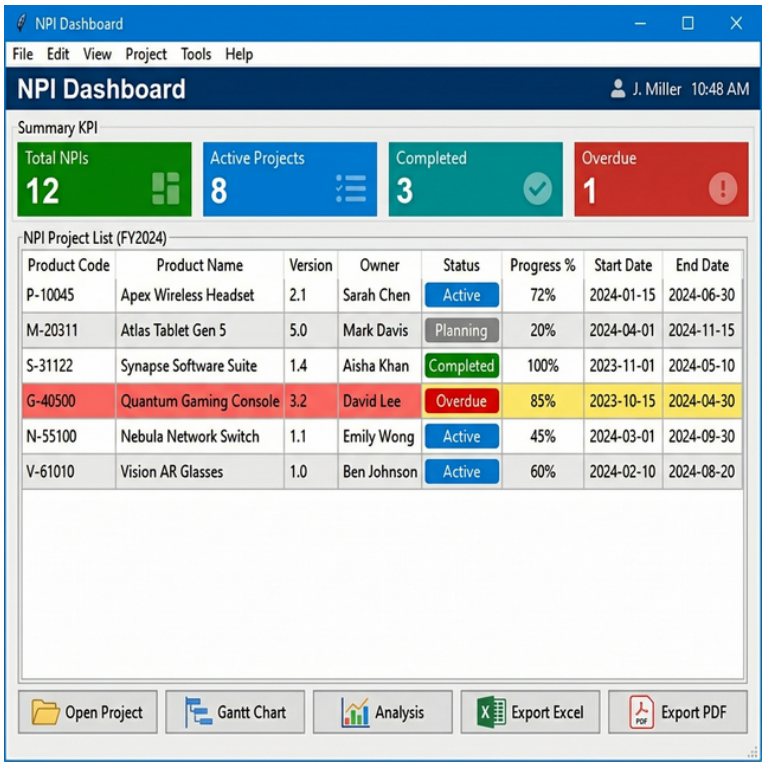

Figure 2 — NPI Dashboard with project list and KPI statistics

3.1 Project Overview & Statistics

The top section displays four KPI cards:

- **Total** — Total number of NPI projects
- **Active** — Projects currently in progress
- **Completed** — Projects with status 'Chiuso' (closed)
- **Overdue** — Projects past their deadline

The main grid shows each project with: Product Code, Product Name, Version, Owner, Status, Progress percentage (calculated from completed tasks), Start Date, and End Date. Rows are color-coded: **red** for overdue, **green** for completed.

3.2 Context Menu Actions

Action	Description
Open Project	Opens the full project detail window for task management
Gantt Chart	Opens the interactive Gantt chart for the selected project
Analysis	Opens the overdue task analysis with email reminder feature
Reopen Project	Reopens a closed project (sets status back to active)

3.3 Export Options

The Dashboard offers multiple export formats:

• **Export Excel (Detailed)** — Full export with task details per project, grouped by customer, with logo and formatting

• **Export Excel (Overview)** — Summary report without task-level detail

• **Export PDF** — Summary PDF with project sections and progress bars

4. Project Management

The Project Window is the heart of the NPI system. Here you manage all tasks, assign owners, track progress, manage dependencies, and attach documents.

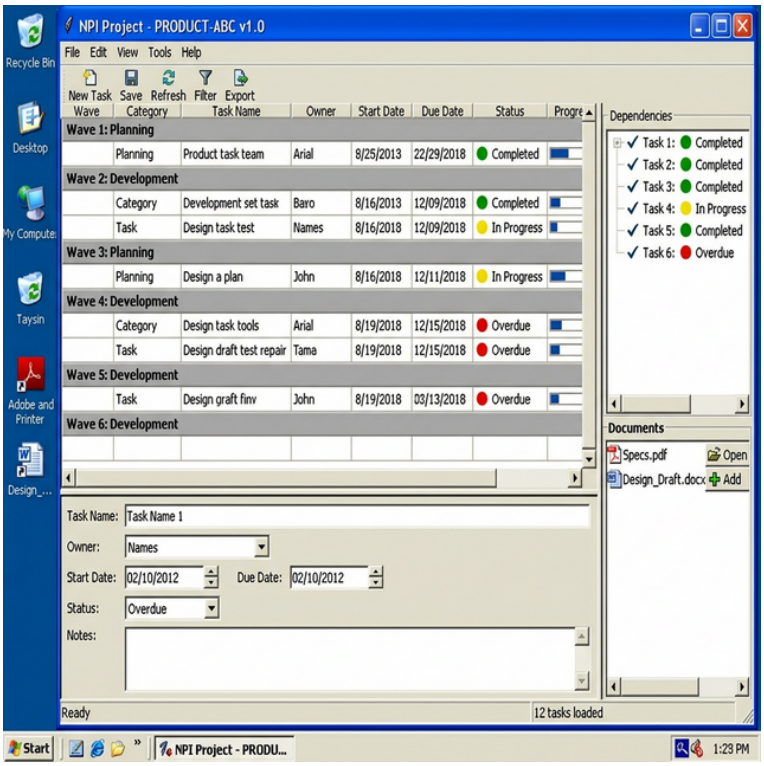

Figure 3 — Project Detail Window with task grid and editing panel

4.1 Task Grid & Editing

The task grid displays all tasks with columns: Wave, Category, Task Name, Owner, Start Date, Due Date, Status, and Progress. Tasks are sorted using topological ordering (respecting dependencies) by default.

Editing a Task:

1. Click on a task row to select it
2. The detail panel below shows editable fields
3. Set Owner, Start Date, Due Date, Status, and Notes
4. Click **Save Task** to persist changes

Visual Indicators:

• Overdue tasks **blink in red** to draw attention
• Completed tasks show with green background
• Tasks with dependencies show a chain link icon

4.2 Task Dependencies

Dependencies define the execution order between tasks. A task with dependencies cannot start until all its predecessors are completed.

Managing Dependencies:
1. Select a task in the grid
2. Click **Dependencies** button to open the dependency window
3. Select a predecessor task from the dropdown
4. Click **Add** to establish the dependency
5. Select an existing dependency and **Remove** to delete it

Tip: Dependencies are shown as arrows in the Gantt chart. Tasks on the "critical path" (blocking other tasks) are highlighted in Risk Alert emails.

4.3 Documents & Attachments

Both projects and individual tasks support document attachments. Documents are stored in a SharePoint-like structure and can be downloaded or replaced.

4.4 Project Hierarchy (Parent-Child)

Complex products may require multiple NPI projects (e.g., variants, sub-assemblies). The hierarchy feature allows you to link projects as parent-child relationships.

Steps:
1. Open the parent project
2. Click **Hierarchy** in the toolbar
3. In the hierarchy window, select available projects and click **Add Child**
4. Child projects appear under the parent in the consolidated Gantt view

4.5 Import Task Assignments

When creating a new NPI project for a similar product, you can import task assignments (owners) from an existing project. This copies the Owner assignments without duplicating the tasks themselves.

4.6 Sort Configuration

The task grid supports multi-column sorting. Click any column header to sort, or use the **Sort Config** button to define a persistent multi-field default sort order saved in a local JSON file.

5. Gantt Chart

The Gantt Chart provides a visual timeline of all project tasks. Built with Plotly, the chart is interactive: hover for details, zoom, and pan.

Figure 4 — Interactive Gantt Chart with color-coded categories and dependencies

5.1 Standard View

The standard view shows all tasks of the current project on a single timeline. Tasks are grouped by category with color-coding. Key features:

- **Red bars** = overdue tasks
- **Dependency arrows** connect linked tasks
- **Vertical red line** = today's date
- **Delay labels** show days overdue on affected tasks
- **Milestones** are marked as diamond shapes

5.2 Consolidated View

For projects with child projects (hierarchy), the consolidated view shows all related projects in a single chart. Each child project is displayed as a collapsible section. Interactive JavaScript features allow expanding/collapsing project sections in the HTML view.

5.3 Export (Excel & PDF)

The Gantt chart can be exported in two formats:

Format	Contents

Excel Export	Multi-sheet workbook: Summary, Gantt Table, Visual Timeline, Statistics, Critical Tasks
PDF Export	Multi-page report: Summary page, Gantt chart (grouped by category), Statistics page

6. Analysis & Reminders

The Analysis window provides a focused view of overdue tasks grouped by owner. From here, you can manually send email reminders to selected task owners.

Steps:
1. Open a project from the Dashboard
2. Right-click and select **Analysis**
3. The window shows each owner with their count of overdue tasks
4. Use checkboxes to select owners
5. Click **Send Reminders** to send email notifications
6. The log panel shows delivery status for each recipient

Tip: This is a manual reminder function, separate from the automated notification system. Use it for urgent follow-ups or ad-hoc escalations.

7. Automatic Notifications

The NPI system includes a background notification service that automatically sends email alerts based on task and project status. The service runs as a daemon thread within the application.

7.1 Scheduling & Configuration

The service checks for notifications at a configured time (default: **14:00**) every working day, respecting a minimum interval between sends.

Parameter	Default	Description
check_time	14:00	Time of day when the system checks for notifications
send_every_n_working_days	2	Minimum working days between notification cycles
include_logo	true	Include company logo in emails
enabled	true	Enable/disable the entire notification service

7.2 Email Types & Recipients

The notification system sends **3 types of email**:

Email Type	Description	Recipients
Unified Person Email	Each person receives 1 consolidated email with 2 sections: - YOUR TASKS: overdue/due-tomorrow tasks assigned to them - YOUR PROJECTS: all alert tasks in projects they own	Task/Project Owners
Project Delay Alert	Sent when projects exceed their deadline or have potential delays. Includes Excel attachment with details.	Project Owners
Weekly Global Report	Sent every non-holiday Monday with a portfolio overview including KPI cards and per-owner breakdown.	System_npi_global_view

Warning: Only subjects with Type = 'Internal' receive automatic notifications. Customers and suppliers are excluded by default. This behavior can be changed in **npi_notifications_config.json** under *recipient_types*.